

GVPT Data Science Camp

Data Transformation

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- ▶ Understanding basic operations in R

Learning Objectives

- ▶ Understanding basic operations in R
- ▶ Introduction to `dplyr`

Learning Objectives

- ▶ Understanding basic operations in R
- ▶ Introduction to `dplyr`
- ▶ Cleaning and transforming data

- ▶ Create new objects with `<-`

```
x <- 6 * 8
```

```
x
```

```
x <- 2 * 15
```

```
x
```

- ▶ Several functions come pre-installed:

```
seq(15, 30)
```

- ▶ Several functions come pre-installed:

```
seq(15, 30)
```

- ▶ You can use functions to create objects:

```
x <- seq(15, 30)
```

```
x
```

What is Data Transformation?

- ▶ Data rarely arrive in exactly the form you need for analysis

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- ▶ Data transformation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries

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- ▶ Data transformation is reshaping data to answer your question: picking out rows and columns, creating new variables, and computing summaries
- ▶ Today's tool: the `dplyr` package from the tidyverse

- ▶ Load the packages we used yesterday:

```
library(tidyverse)  
library(readxl)
```

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```
library(tidyverse)
library(readxl)
```

- ▶ Re-run yesterday's setup code (`read_excel()`, `rename()`, `select()`, and `mutate()`) to re-create the `cel` object, then view the data:

```
view(cel)
```

► In `cel`:

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- ▶ `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)

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- ▶ In `cel`:
- ▶ `fct` stands for factors, which `R` uses to represent categorical variables with fixed values — like `benchmark` (Below, Meets, Exceeds)
- ▶ `dbl` stands for doubles — like `les` and `ideology`
- ▶ `chr` stands for strings (character vectors) — like `member`, `state`, and `party`

► `int` stands for integer

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- ▶ `dtm` stands for date-times (a date + time)

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- ▶ `dtm` stands for date-times (a date + time)
- ▶ `lgl` stands for logical, vectors that contain only `TRUE` or `FALSE`

- ▶ Five basic functions:

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- ▶ `filter()`

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- ▶ Five basic functions:
- ▶ `filter()`
- ▶ `arrange()`
- ▶ `select()`
- ▶ `mutate()`
- ▶ `summarise()`

Filtering Rows

```
filter(cel, state == "MD", congress > 110)
```

	member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	bills_passed	benchmark	party
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1	Bartlett, Roscoe	111	2009	MD	0	0	58	0.493		9	1926	0.172	0	Below	Republican
2	Cummings, Elijah	111	2009	MD	1	0	80	-0.438		7	1951	1.74	5	Meets	Democrat
3	Edwards, Donna	111	2009	MD	1	1	86	-0.562		1	1958	0.101	0	Below	Democrat
4	Hoyer, Steny	111	2009	MD	1	0	74	-0.380		15	1939	1.04	3	Meets	Democrat
5	Kratovil, Frank	111	2009	MD	1	0	49	-0.0370		1	1968	1.56	2	Exceeds	Democrat
6	Ruppersberger, C.	111	2009	MD	1	0	72	-0.295		4	1946	0.720	1	Meets	Democrat
7	Sarbanes, John	111	2009	MD	1	0	70	-0.468		2	1962	1.95	4	Exceeds	Democrat
8	Van Hollen, Chris	111	2009	MD	1	0	75	-0.391		4	1959	1.61	2	Exceeds	Democrat
9	Bartlett, Roscoe	112	2011	MD	0	0	61	0.493		10	1926	0.107	0	Below	Republican
10	Cummings, Elijah	112	2011	MD	1	0	75	-0.438		8	1951	0.443	0	Meets	Democrat

i 54 more rows
i Use `print(n = ...)` to see more rows

- ▶ First argument must always be a data frame

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- ▶ The arguments that follow describe which columns to operate on, using the variable names (without quotes)

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- ▶ The arguments that follow describe which columns to operate on, using the variable names (without quotes)
- ▶ Output is always a new data frame

Filtering Rows

```
filter(cel, state %in% c("MD", "VA"))
```

member	congress	year	state	dem	female	votepct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1 Bauman, Robert	93	1973	MD	0	0	NA	0.533	1	1937	0.0620	0	Below	Republican
2 Broyhill, Joel	93	1973	VA	0	0	56	0.159	11	1919	2.42	6	Exceeds	Republican
3 Butler, M.	93	1973	VA	0	0	55	0.332	1	1925	0.0114	0	Below	Republican
4 Byron, Goodloe	93	1973	MD	1	0	65	-0.0140	2	1929	0.124	0	Below	Democrat
5 Daniel, Robert	93	1973	VA	0	0	47	0.318	1	1936	0.0227	0	Below	Republican
6 Daniel, W.	93	1973	VA	1	0	100	0.161	3	1914	0.0382	0	Below	Democrat
7 Downing, Thomas	93	1973	VA	1	0	78	-0.0650	8	1919	1.94	4	Meets	Democrat
8 Gude, Gilbert	93	1973	MD	0	0	64	-0.0280	4	1923	1.46	3	Exceeds	Republican
9 Hogan, Lawrence	93	1973	MD	0	0	63	0.158	3	1928	0.200	0	Meets	Republican
10 Holt, Marjorie	93	1973	MD	0	1	59	0.368	1	1920	0.222	1	Meets	Republican
% # 482 more rows													
% # Use `print(n = ...)` to see more rows													

Filtering Rows

```
filter(cel, vote_pct > 50 & vote_pct < 60)
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Abdnor, James	93	1973	SD	0	0	55	0.241	1	1923	0.110	0	Below	Republican
2 Abzug, Bella	93	1973	NY	1	1	56	-0.597	2	1920	0.762	1	Meets	Democrat
3 Annunzio, Frank	93	1973	IL	1	0	53	-0.398	5	1915	0.628	2	Below	Democrat
4 Arends, Leslie	93	1973	IL	0	0	57	0.301	20	1895	0.0155	0	Below	Republican
5 Ashbrook, John	93	1973	OH	0	0	57	0.612	7	1928	0.172	0	Below	Republican
6 Baker, LaMar	93	1973	TN	0	0	55	0.308	2	1915	0.819	1	Exceeds	Republican
7 Beard, Robin	93	1973	TN	0	0	55	0.297	1	1939	0.0475	0	Below	Republican
8 Bergland, Bob	93	1973	MN	1	0	59	-0.380	2	1928	0.125	0	Below	Democrat
9 Brademas, John	93	1973	IN	1	0	55	-0.462	8	1927	5.62	7	Exceeds	Democrat
10 Breckinridge, John	93	1973	KY	1	0	52	-0.223	1	1913	0.484	1	Meets	Democrat

i 3,013 more rows
i Use `print(n = ...)` to see more rows

Filtering Rows

```
filter(cel, vote_pct > 50 | vote_pct < 60)
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Abdnor, James	93	1973	SD	0	0	55	0.241	1	1923	0.110	0	Below	Republican
2 Abzug, Bella	93	1973	NY	1	1	56	-0.597	2	1920	0.762	1	Meets	Democrat
3 Adams, Brock	93	1973	WA	1	0	85	-0.396	5	1927	1.24	2	Meets	Democrat
4 Addabbo, Joseph	93	1973	NY	1	0	75	-0.402	7	1925	0.155	0	Below	Democrat
5 Albert, Carl	93	1973	OK	1	0	93	-0.392	14	1908	0.00103	0	Below	Democrat
6 Alexander, Bill	93	1973	AR	1	0	100	-0.410	3	1934	1.88	2	Meets	Democrat
7 Anderson, John	93	1973	IL	0	0	72	0.183	7	1922	0.289	0	Meets	Republican
8 Anderson, Glenn	93	1973	CA	1	0	75	-0.269	3	1913	0.447	0	Meets	Democrat
9 Andrews, Ike	93	1973	NC	1	0	50	-0.178	1	1925	0.0879	0	Below	Democrat
10 Andrews, Mark	93	1973	ND	0	0	73	0.0870	6	1926	0.106	0	Below	Republican

i 11,236 more rows
i Use `print(n = ...)` to see more rows

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)

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- ▶ Find all members from Maryland

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- ▶ Find all members from Maryland
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- ▶ Find all member-Congress observations with a vote share greater than 60% and less than 70%

- ▶ Find all member-Congress observations where the member ran unopposed (received 100% of the vote)
- ▶ Find all members from Maryland
- ▶ Find all members from Maryland and Virginia
- ▶ Find all member-Congress observations with a vote share greater than 60% and less than 70%
- ▶ Find all member-Congress observations with a vote share less than 60% or greater than 70%

Check your Answers

```
filter(cel, vote_pct == 100)
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Alexander, Bill	93	1973	AR	1	0	100	-0.410	3	1934	1.88	2	Meets	Democrat
2 Boland, Edward	93	1973	MA	1	0	100	-0.306	11	1911	2.05	2	Meets	Democrat
3 Breaux, John	93	1973	LA	1	0	100	-0.123	1	1944	1.16	1	Exceeds	Democrat
4 Brinkley, Jack	93	1973	GA	1	0	100	-0.0730	4	1930	0.217	0	Below	Democrat
5 Burke, James	93	1973	MA	1	0	100	-0.400	8	1910	0.789	0	Meets	Democrat
6 Burleson, Omar	93	1973	TX	1	0	100	0.00200	14	1906	0.130	0	Below	Democrat
7 Conte, Silvio	93	1973	MA	0	0	100	-0.0230	8	1921	1.46	3	Exceeds	Republican
8 Daniel, W.	93	1973	VA	1	0	100	0.161	3	1914	0.0382	0	Below	Democrat
9 Donohue, Harold	93	1973	MA	1	0	100	-0.296	14	1901	1.42	3	Meets	Democrat
10 Duncan, John	93	1973	TN	0	0	100	0.194	5	1919	0.257	0	Meets	Republican

i 740 more rows
i Use `print(n = ...)` to see more rows

Check your Answers

```
filter(cel, state == "MD")
```

member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	ills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1 Bauman, Robert	93	1973	MD	0	0	NA	0.533		1	1937	0.062	0	Below	Republican
2 Byron, Goodloe	93	1973	MD	1	0	65	-0.014		2	1929	0.124	0	Below	Democrat
3 Gude, Gilbert	93	1973	MD	0	0	64	-0.028		4	1923	1.46	3	Exceeds	Republican
4 Hogan, Lawrence	93	1973	MD	0	0	63	0.158		3	1928	0.200	0	Meets	Republican
5 Holt, Marjorie	93	1973	MD	0	1	59	0.368		1	1920	0.222	1	Meets	Republican
6 Long, Clarence	93	1973	MD	1	0	66	-0.274		6	1908	0.210	0	Below	Democrat
7 Mills, William	93	1973	MD	0	0	NA	0.210		2	1924	0.051	0	Below	Republican
8 Mitchell, Parren	93	1973	MD	1	0	80	-0.526		2	1922	0.092	0	Below	Democrat
9 Sarbanes, Paul	93	1973	MD	1	0	70	-0.435		2	1933	0.010	0	Below	Democrat
10 Bauman, Robert	94	1975	MD	0	0	53	0.533		2	1937	0.080	0	Below	Republican

i 201 more rows
i Use `print(n = ...)` to see more rows

Check your Answers

```
filter(cel, state %in% c("MD", "VA"))
```

member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Bauman, Robert	93	1973	MD	0	0	NA	0.533		1	1937	0.0620		0 Below	Republican
2 Broyhill, Joel	93	1973	VA	0	0	56	0.159		11	1919	2.42		6 Exceeds	Republican
3 Butler, M.	93	1973	VA	0	0	55	0.332		1	1925	0.0114		0 Below	Republican
4 Byron, Goodloe	93	1973	MD	1	0	65	-0.0140		2	1929	0.124		0 Below	Democrat
5 Daniel, Robert	93	1973	VA	0	0	47	0.318		1	1936	0.0227		0 Below	Republican
6 Daniel, W.	93	1973	VA	1	0	100	0.161		3	1914	0.0382		0 Below	Democrat
7 Downing, Thomas	93	1973	VA	1	0	78	-0.0650		8	1919	1.94		4 Meets	Democrat
8 Gude, Gilbert	93	1973	MD	0	0	64	-0.0280		4	1923	1.46		3 Exceeds	Republican
9 Hogan, Lawrence	93	1973	MD	0	0	63	0.158		3	1928	0.200		0 Meets	Republican
10 Holt, Marjorie	93	1973	MD	0	1	59	0.368		1	1920	0.222		1 Meets	Republican

i 482 more rows
i Use `print(n = ...)` to see more rows

Check your Answers

```
filter(cel, vote_pct > 60 & vote_pct < 70)
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	bills_passed	benchmark	party	
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>		<dbl>	<fct>	<chr>
1 Armstrong, William	93	1973	CO	0	0	62	0.508	1	1937	0.137		1 Meets	Republican	
2 Ashley, Thomas	93	1973	OH	1	0	69	-0.350	10	1923	3.76		5 Exceeds	Democrat	
3 Aspin, Les	93	1973	WI	1	0	64	-0.320	2	1938	0.472		0 Meets	Democrat	
4 Bafalis, L.	93	1973	FL	0	0	62	0.303	1	1929	0.0517		0 Below	Republican	
5 Barrett, William	93	1973	PA	1	0	66	-0.469	13	1896	1.65		2 Meets	Democrat	
6 Bell, Alphonzo	93	1973	CA	0	0	61	0.183	7	1914	0.233		0 Meets	Republican	
7 Biester, Edward	93	1973	PA	0	0	64	0.0350	4	1931	0.115		0 Below	Republican	
8 Bolling, Richard	93	1973	MO	1	0	63	-0.504	13	1916	0		0 Below	Democrat	
9 Bowen, David	93	1973	MS	1	0	62	-0.134	1	1932	0.588		1 Meets	Democrat	
10 Brasco, Frank	93	1973	NY	1	0	64	-0.423	4	1932	0.585		1 Meets	Democrat	

i 3,413 more rows
i Use `print(n = ...)` to see more rows

Check your Answers

```
filter(cel, vote_pct < 60 | vote_pct > 70)
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	ills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Abdnor, James	93	1973	SD	0	0	55	0.241	1	1923	0.110		0 Below	Republican
2 Abzug, Bella	93	1973	NY	1	1	56	-0.597	2	1920	0.762		1 Meets	Democrat
3 Adams, Brock	93	1973	WA	1	0	85	-0.396	5	1927	1.24		2 Meets	Democrat
4 Addabbo, Joseph	93	1973	NY	1	0	75	-0.402	7	1925	0.155		0 Below	Democrat
5 Albert, Carl	93	1973	OK	1	0	93	-0.392	14	1908	0.00103		0 Below	Democrat
6 Alexander, Bill	93	1973	AR	1	0	100	-0.410	3	1934	1.88		2 Meets	Democrat
7 Anderson, John	93	1973	IL	0	0	72	0.183	7	1922	0.289		0 Meets	Republican
8 Anderson, Glenn	93	1973	CA	1	0	75	-0.269	3	1913	0.447		0 Meets	Democrat
9 Andrews, Ike	93	1973	NC	1	0	50	-0.178	1	1925	0.0879		0 Below	Democrat
10 Andrews, Mark	93	1973	ND	0	0	73	0.0870	6	1926	0.106		0 Below	Republican

i 7,132 more rows
i Use `print(n = ...)` to see more rows

► `==` is equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to
- ▶ `>=` is greater than or equal to

Important Features

- ▶ `==` is equal to
- ▶ `!=` is not equal to
- ▶ `>=` is greater than or equal to
- ▶ `<=` is less than or equal to

▶ | is or

Important Features

- ▶ `|` is or
- ▶ `&` is AND

Important Features

- ▶ `|` is or
- ▶ `&` is AND
- ▶ `%in%` is in

Arrange Rows with arrange()

```
arrange(cel, member, congress)
```

	member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	bills_passed	benchmark	party
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1	Abdnor, James	93	1973	SD	0	0	55	0.241		1	1923	0.110		0 Below	Republican
2	Abdnor, James	94	1975	SD	0	0	68	0.241		2	1923	0.186		0 Meets	Republican
3	Abdnor, James	95	1977	SD	0	0	70	0.241		3	1923	0.428		1 Meets	Republican
4	Abdnor, James	96	1979	SD	0	0	NA	0.241		4	1923	0.313		1 Meets	Republican
5	Abercrombie, Neil	99	1985	HI	1	0	NA	-0.432		1	1938	0.0154		0 Below	Democrat
6	Abercrombie, Neil	102	1991	HI	1	0	60	-0.432		1	1938	1.10		0 Exceeds	Democrat
7	Abercrombie, Neil	103	1993	HI	1	0	73	-0.432		2	1938	0.0982		0 Below	Democrat
8	Abercrombie, Neil	104	1995	HI	1	0	54	-0.432		3	1938	0.693		1 Exceeds	Democrat
9	Abercrombie, Neil	105	1997	HI	1	0	50	-0.432		4	1938	0.235		0 Meets	Democrat
10	Abercrombie, Neil	106	1999	HI	1	0	62	-0.432		5	1938	0.427		1 Meets	Democrat

i 11,596 more rows
i Use `print(n = ...)` to see more rows

Arrange Rows with arrange()

```
arrange(cel, member, desc(congress))
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>	dem <dbl>	female <dbl>	vote_pct <dbl>	ideology <dbl>	seniority <dbl>	born <dbl>	les <dbl>	bills_passed <dbl>	benchmark <fct>	party <chr>
1	Abdnor, James	96	1979	SD	0	0	NA	0.241	4	1923	0.313	1	Meets	Republican
2	Abdnor, James	95	1977	SD	0	0	70	0.241	3	1923	0.428	1	Meets	Republican
3	Abdnor, James	94	1975	SD	0	0	68	0.241	2	1923	0.186	0	Meets	Republican
4	Abdnor, James	93	1973	SD	0	0	55	0.241	1	1923	0.110	0	Below	Republican
5	Abercrombie, Neil	111	2009	HI	1	0	77	-0.432	10	1938	0.674	1	Below	Democrat
6	Abercrombie, Neil	110	2007	HI	1	0	69	-0.432	9	1938	0.939	3	Meets	Democrat
7	Abercrombie, Neil	109	2005	HI	1	0	63	-0.432	8	1938	0.311	0	Meets	Democrat
8	Abercrombie, Neil	108	2003	HI	1	0	73	-0.432	7	1938	0.353	0	Meets	Democrat
9	Abercrombie, Neil	107	2001	HI	1	0	69	-0.432	6	1938	0.390	0	Meets	Democrat
10	Abercrombie, Neil	106	1999	HI	1	0	62	-0.432	5	1938	0.427	1	Meets	Democrat

i 11,596 more rows
i Use `print(n = ...)` to see more rows

- ▶ Which member-Congress observations have the highest Legislative Effectiveness Score (**les**)?

- ▶ Which member-Congress observations have the highest Legislative Effectiveness Score (`les`)?
- ▶ Which member-Congress observation passed the most bills through the House (`bills_passed`)?

Check your Answers

```
arrange(cel, desc(les))
```

member	congress	year	state	dem	female	vote_pct	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1 Rangel, Charles	110	2007	NY	1	0	94	-0.514	19	1930	18.7	33	Exceeds	Democrat
2 Murphy, John	96	1979	NY	1	0	54	-0.406	9	1926	17.9	31	Exceeds	Democrat
3 Sullivan, Leonor	93	1973	MO	1	1	69	-0.386	11	1902	17.6	28	Exceeds	Democrat
4 Sensenbrenner, F.	109	2005	WI	0	0	67	0.638	14	1943	17.6	25	Exceeds	Republican
5 Murphy, John	95	1977	NY	1	0	NA	-0.406	8	1926	17.2	31	Exceeds	Democrat
6 Ullman, Al	96	1979	OR	1	0	69	-0.367	12	1914	16.4	18	Exceeds	Democrat
7 Smith, Lamar	112	2011	TX	0	0	69	0.425	13	1947	16.3	20	Exceeds	Republican
8 Young, Don	108	2003	AK	0	0	75	0.284	16	1933	16.3	20	Exceeds	Republican
9 Young, Don	107	2001	AK	0	0	70	0.284	15	1933	15.8	22	Exceeds	Republican
10 Young, Don	109	2005	AK	0	0	71	0.284	17	1933	15.3	18	Exceeds	Republican

i 11,596 more rows
i Use `print(n = ...)` to see more rows
i

Check your Answers

```
arrange(cel, desc(bills_passed))
```

member	congress	year	state	dem	female	votept	ideology	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Rangel, Charles	110	2007	NY	1	0	94	-0.514	19	1930	18.7	33	Exceeds	Democrat
2 Murphy, John	95	1977	NY	1	0	NA	-0.406	8	1926	17.2	31	Exceeds	Democrat
3 Murphy, John	96	1979	NY	1	0	54	-0.406	9	1926	17.9	31	Exceeds	Democrat
4 Sullivan, Leonor	93	1973	MO	1	1	69	-0.386	11	1902	17.6	28	Exceeds	Democrat
5 Sullivan, Leonor	94	1975	MO	1	1	75	-0.386	12	1902	12.7	25	Exceeds	Democrat
6 Sensenbrenner, F.	109	2005	WI	0	0	67	0.638	14	1943	17.6	25	Exceeds	Republican
7 Young, Don	107	2001	AK	0	0	70	0.284	15	1933	15.8	22	Exceeds	Republican
8 Miller, George	110	2007	CA	1	0	84	-0.552	17	1945	10.6	22	Exceeds	Democrat
9 Oberstar, James	110	2007	MN	1	0	64	-0.553	17	1934	13.0	22	Exceeds	Democrat
10 Rogers, Paul	93	1973	FL	1	0	60	-0.0850	10	1921	11.5	21	Exceeds	Democrat

i 11,596 more rows
i Use `print(n = ...)` to see more rows

Select Columns with select()

```
select(cel, member, congress, les)
```

```
      member      congress      les  
      <chr>      <dbl>      <dbl>  
1 Abdnor, James      93 0.110  
2 Abzug, Bella      93 0.762  
3 Adams, Brock      93 1.24  
4 Addabbo, Joseph   93 0.155  
5 Albert, Carl      93 0.00103  
6 Alexander, Bill   93 1.88  
7 Anderson, John    93 0.289  
8 Anderson, Glenn   93 0.447  
9 Andrews, Ike      93 0.0879  
10 Andrews, Mark    93 0.106  
# i 11,596 more rows  
# i Use `print(n = ...)` to see more rows  
~ |
```

Select Columns with select()

```
select(cel, member:state)
```

	member <chr>	congress <dbl>	year <dbl>	state <chr>
1	Abdnor, James	93	1973	SD
2	Abzug, Bella	93	1973	NY
3	Adams, Brock	93	1973	WA
4	Addabbo, Joseph	93	1973	NY
5	Albert, Carl	93	1973	OK
6	Alexander, Bill	93	1973	AR
7	Anderson, John	93	1973	IL
8	Anderson, Glenn	93	1973	CA
9	Andrews, Ike	93	1973	NC
10	Andrews, Mark	93	1973	ND

i 11,596 more rows
i Use `print(n = ...)`` to see more rows
,

Select Columns with select()

```
select(cel, -(vote_pct:ideology))
```

member	congress	year	state	dem	female	seniority	born	les	bills_passed	benchmark	party
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<fct>	<chr>
1 Abdnor, James	93	1973	SD	0	0	1	1923	0.110	0	Below	Republican
2 Abzug, Bella	93	1973	NY	1	1	2	1920	0.762	1	Meets	Democrat
3 Adams, Brock	93	1973	WA	1	0	5	1927	1.24	2	Meets	Democrat
4 Addabbo, Joseph	93	1973	NY	1	0	7	1925	0.155	0	Below	Democrat
5 Albert, Carl	93	1973	OK	1	0	14	1908	0.00103	0	Below	Democrat
6 Alexander, Bill	93	1973	AR	1	0	3	1934	1.88	2	Meets	Democrat
7 Anderson, John	93	1973	IL	0	0	7	1922	0.289	0	Meets	Republican
8 Anderson, Glenn	93	1973	CA	1	0	3	1913	0.447	0	Meets	Democrat
9 Andrews, Ike	93	1973	NC	1	0	1	1925	0.0879	0	Below	Democrat
10 Andrews, Mark	93	1973	ND	0	0	6	1926	0.106	0	Below	Republican
# i 11,596 more rows											
# i Use `print(n = ...)` to see more rows											

- ▶ Select only the `member`, `congress`, and `votepct` variables from `cel`

Exercise

- ▶ Select only the `member`, `congress`, and `votepct` variables from `cel`
- ▶ What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "votepct")
```

- ▶ Select only the `member`, `congress`, and `vote_pct` variables from `cel`
- ▶ What does the `any_of()` function do? Why might it be useful with this vector?

```
vars <- c("member", "congress", "vote_pct")
```

- ▶ What does this code produce?

```
select(cel, starts_with("b"))
```

Check your Answers

```
select(cel, member, congress, vote_pct)
```

	member <chr>	congress <dbl>	vote_pct <dbl>
1	Abdnor, James	93	55
2	Abzug, Bella	93	56
3	Adams, Brock	93	85
4	Addabbo, Joseph	93	75
5	Albert, Carl	93	93
6	Alexander, Bill	93	100
7	Anderson, John	93	72
8	Anderson, Glenn	93	75
9	Andrews, Ike	93	50
10	Andrews, Mark	93	73

```
# i 11,596 more rows  
# i Use `print(n = ...)` to see more rows  
i
```

Check your Answers

`any_of()` selects any of the named columns that exist, without erroring on ones that don't:

```
vars <- c("member", "congress", "votepct")  
  
select(cel, any_of(vars))
```

	member <chr>	congress <dbl>	votepct <dbl>
1	Abdnor, James	93	55
2	Abzug, Bella	93	56
3	Adams, Brock	93	85
4	Addabbo, Joseph	93	75
5	Albert, Carl	93	93
6	Alexander, Bill	93	100
7	Anderson, John	93	72
8	Anderson, Glenn	93	75
9	Andrews, Ike	93	50
10	Andrews, Mark	93	73

11,596 more rows

Check your Answers

All columns whose names start with "b": **born**, **bills_passed**, and **benchmark**

```
select(cel, starts_with("b"))
```

```
   born bills_passed benchmark
   <dbl>         <dbl> <fct>
1  1923             0 Below
2  1920             1 Meets
3  1927             2 Meets
4  1925             0 Below
5  1908             0 Below
6  1934             2 Meets
7  1922             0 Meets
8  1913             0 Meets
9  1925             0 Below
10 1926             0 Below
# i 11,596 more rows
# i Use `print(n = ...)` to see more rows
. |
```

Add New Variables with mutate()

```
mutate(cel, age = year - born)
```

	member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	bill	passed	benchmark	party	age
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	Abdnor, James	93	1973	SD	0	0	55	0.241		1	1923	0.110	0	Below		Republican	50
2	Abzug, Bella	93	1973	NY	1	1	56	-0.597		2	1920	0.762	1	Meets		Democrat	53
3	Adams, Brock	93	1973	WA	1	0	85	-0.396		5	1927	1.24	2	Meets		Democrat	46
4	Addabbo, Joseph	93	1973	NY	1	0	75	-0.402		7	1925	0.155	0	Below		Democrat	48
5	Albert, Carl	93	1973	OK	1	0	93	-0.392		14	1908	0.00103	0	Below		Democrat	65
6	Alexander, Bill	93	1973	AR	1	0	100	-0.410		3	1934	1.88	2	Meets		Democrat	39
7	Anderson, John	93	1973	IL	0	0	72	0.183		7	1922	0.289	0	Meets		Republican	51
8	Anderson, Glenn	93	1973	CA	1	0	75	-0.269		3	1913	0.447	0	Meets		Democrat	60
9	Andrews, Ike	93	1973	NC	1	0	50	-0.178		1	1925	0.0879	0	Below		Democrat	48
10	Andrews, Mark	93	1973	ND	0	0	73	0.0870		6	1926	0.106	0	Below		Republican	47

i 11,596 more rows
i Use `print(n = ...)` to see more rows

Select, Transform, and Add New Variables with transmute()

```
transmute(cel, member, congress, age = year - born)
```

	member <chr>	congress <dbl>	age <dbl>
1	Abdnor, James	93	50
2	Abzug, Bella	93	53
3	Adams, Brock	93	46
4	Addabbo, Joseph	93	48
5	Albert, Carl	93	65
6	Alexander, Bill	93	39
7	Anderson, John	93	51
8	Anderson, Glenn	93	60
9	Andrews, Ike	93	48
10	Andrews, Mark	93	47

```
# i 11,596 more rows  
# i Use `print(n = ...)` to see more rows  
.
```

Create Summaries with summarise()

```
summarise(  
  cel, avg_bills = mean(bills_passed), avg_seniority = mean(seniority)  
)
```

	avg_bills	avg_seniority
	<dbl>	<dbl>
1	1.48	5.27
.		

Creating Grouped Summaries with `group_by()` and `summarise()`

```
cel_party <- group_by(cel, party)

summarise(
  cel_party, avg_bills = mean(bills_passed), avg_seniority =
    mean(seniority)
)
```

	party	avg_bills	avg_seniority
	<chr>	<dbl>	<dbl>
1	Democrat	1.57	5.75
2	Republican	1.38	4.72
.	1		

- ▶ Calculate each member's age in each Congress

- ▶ Calculate each member's age in each Congress
- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset

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- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset
- ▶ Find the member with the lowest average bills passed across these Congresses

- ▶ Calculate each member's age in each Congress
- ▶ Find each member's average number of bills passed across all Congresses in the `cel` dataset
- ▶ Find the member with the lowest average bills passed across these Congresses
- ▶ Find the member with the highest average bills passed across these Congresses

Check your Answers

```
mutate(cel, age = year - born)
```

	member	congress	year	state	dem	female	vote	pct	ideology	seniority	born	les	ills_passed	benchmark	party	age
	<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	Abdnor, James	93	1973	SD	0	0	55	0.241		1	1923	0.110		0 Below	Republican	50
2	Abzug, Bella	93	1973	NY	1	1	56	-0.597		2	1920	0.762		1 Meets	Democrat	53
3	Adams, Brock	93	1973	WA	1	0	85	-0.396		5	1927	1.24		2 Meets	Democrat	46
4	Addabbo, Joseph	93	1973	NY	1	0	75	-0.402		7	1925	0.155		0 Below	Democrat	48
5	Albert, Carl	93	1973	OK	1	0	93	-0.392		14	1908	0.00103		0 Below	Democrat	65
6	Alexander, Bill	93	1973	AR	1	0	100	-0.410		3	1934	1.88		2 Meets	Democrat	39
7	Anderson, John	93	1973	IL	0	0	72	0.183		7	1922	0.289		0 Meets	Republican	51
8	Anderson, Glenn	93	1973	CA	1	0	75	-0.269		3	1913	0.447		0 Meets	Democrat	60
9	Andrews, Ike	93	1973	NC	1	0	50	-0.178		1	1925	0.0879		0 Below	Democrat	48
10	Andrews, Mark	93	1973	ND	0	0	73	0.0870		6	1926	0.106		0 Below	Republican	47

i 11,596 more rows
i Use `print(n = ...)` to see more rows

Check your Answers

```
cel_member <- group_by(cel, member)

avg_bills_by_member <- summarise(cel_member, avg_bills =
  mean(bills_passed))

avg_bills_by_member
```

	member	avg_bills
	<chr>	<dbl>
1	Abdnor, James	0.5
2	Abercrombie, Neil	0.545
3	Abraham, Ralph	1.33
4	Abzug, Bella	1.5
5	Acevedo-Vila, Anibal	2
6	Ackerman, Gary	1.33
7	Adams, Alma	1.33
8	Adams, Brock	1.5
9	Adams, Sandy	2
10	Addabbo, Joseph	1.57
# i 2,223 more rows		
# i Use `print(n = ...)` to see more rows		

```
arrange(avg_bills_by_member, avg_bills)
```

```
member      avg_bills
<chr>      <dbl>
1 Albert, Carl      0
2 Ammerman, Joseph  0
3 Amo, Gabe         0
4 Arends, Leslie    0
5 Ashbrook, John    0
6 Austria, Steve    0
7 Bacchus, Jim      0
8 Badillo, Herman   0
9 Baesler, Scotty   0
10 Bafalis, L.      0
# i 2,223 more rows
# i Use `print(n = ...)` to see more rows
```

```
arrange(avg_bills_by_member, desc(avg_bills))
```

```
  member      avg_bills  
  <chr>      <dbl>  
1 Sullivan, Leonor    26.5  
2 Rogers, Paul       19.7  
3 Murphy, John       17  
4 Ullman, Al         14.8  
5 Rodino, Peter      12.1  
6 Mahon, George      12  
7 Hays, Wayne        11  
8 Katko, John        10.5  
9 Brooks, Jack       10.2  
10 Jones, Robert      10  
# i 2,223 more rows  
# i Use `print(n = ...)` to see more rows  
!
```

Utilize Multiple Operations with the Pipe

- ▶ We created a lot of different objects during our calculations

Utilize Multiple Operations with the Pipe

- ▶ We created a lot of different objects during our calculations
- ▶ Can we avoid creating unneeded objects?

Utilize Multiple Operations with the Pipe

- ▶ We created a lot of different objects during our calculations
- ▶ Can we avoid creating unneeded objects?
- ▶ We can utilize a feature of the tidyverse: the pipe

Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:

Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)

Utilize Multiple Operations with the Pipe

- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)
- ▶ do this `|>` (and then...)

Utilize Multiple Operations with the Pipe

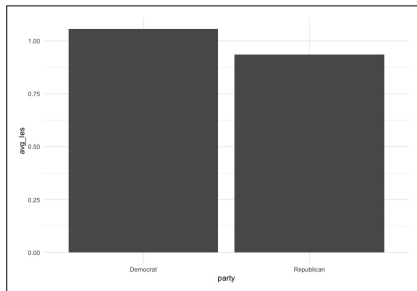
- ▶ Pipe can be read as:
- ▶ Take this `|>` (and then...)
- ▶ do this `|>` (and then...)
- ▶ do this

```
cel |>
  group_by(state) |>
  summarise(avg_les = mean(les)) |>
  arrange(desc(avg_les))
```

```
state avg_les
<chr>  <dbl>
1 AK      4.87
2 DC      2.65
3 OR      1.54
4 MT      1.52
5 UT      1.50
6 NM      1.49
7 NJ      1.40
8 AZ      1.39
9 DE      1.30
10 MS     1.28
# i 46 more rows
# i Use 'print(n = ...)' to see more rows
```

Utilize Multiple Operations with the Pipe

```
cel |>  
  group_by(parties) |>  
  summarise(avg_les = mean(les)) |>  
  ggplot(aes(x = parties, y = avg_les)) +  
  geom_col() +  
  theme_minimal()
```



- ▶ Base pipe:

A Note on the Pipe

- ▶ Base pipe:
- ▶ $|>$

A Note on the Pipe

- ▶ Base pipe:
- ▶ `|>`
- ▶ Can be used without loading any packages

A Note on the Pipe

- ▶ Base pipe:
- ▶ `|>`
- ▶ Can be used without loading any packages
- ▶ Relatively new: introduced in 2021

A Note on the Pipe

- ▶ Tidyverse pipe:

A Note on the Pipe

- ▶ Tidyverse pipe:
- ▶ `%>%`

A Note on the Pipe

- ▶ Tidyverse pipe:
- ▶ `%>%`
- ▶ Must load `dplyr` or `magrittr` to use

- ▶ Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2007 onward). Display only the member and average LES variables

- ▶ Calculate the average Legislative Effectiveness Score for Maryland members serving in Congresses after the 110th (2007 onward). Display only the member and average LES variables
- ▶ Plot the results

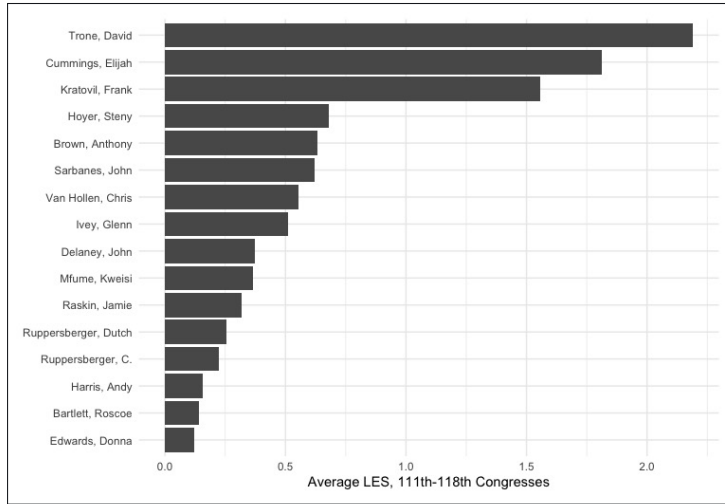
Check your Answers

```
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les))
```

member	avg_les
<chr>	<dbl>
1 Bartlett, Roscoe	0.139
2 Brown, Anthony	0.634
3 Cummings, Elijah	1.81
4 Delaney, John	0.374
5 Edwards, Donna	0.122
6 Harris, Andy	0.156
7 Hoyer, Steny	0.679
8 Ivey, Glenn	0.509
9 Kratovil, Frank	1.56
10 Mfume, Kweisi	0.366
11 Raskin, Jamie	0.317
12 Ruppersberger, C.	0.222
13 Ruppersberger, Dutch	0.254
14 Sarbanes, John	0.622
15 Trone, David	2.19
16 Van Hollen, Chris	0.552

```
cel |>
  filter(state == "MD", congress > 110) |>
  group_by(member) |>
  summarise(avg_les = mean(les)) |>
  ggplot(aes(x = avg_les, y = fct_reorder(member, avg_les))) +
  geom_col() +
  theme_minimal() +
  labs(x = "Average LES, 111th-118th Congresses", y = NULL)
```

Check your Answers



- ▶ Familiarized yourself with basic R syntax

- ▶ Familiarized yourself with basic R syntax
- ▶ Learned how to transform your data

- ▶ Familiarized yourself with basic **R** syntax
- ▶ Learned how to transform your data
- ▶ Wrote concise code that is easy to follow

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:

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- ▶ Is relevant to your research interests

- ▶ On Thursday, you will apply the skills learned in this course to an idea that interests you. In doing so, you will need to find a dataset that:
- ▶ Is relevant to your research interests
- ▶ Contains continuous and categorical variables

Questions?